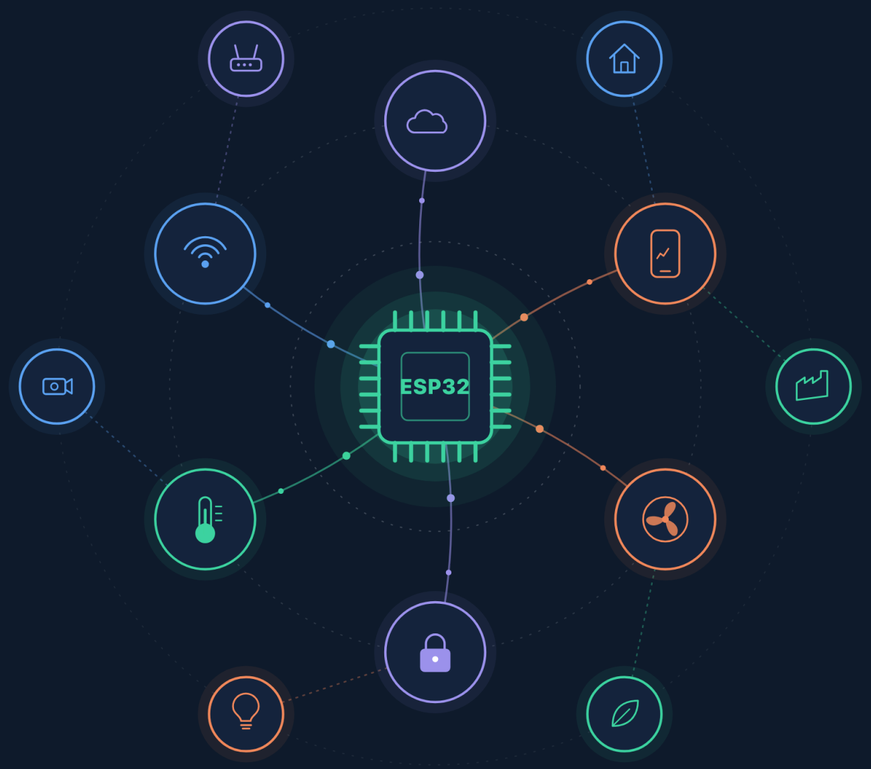
MODULE 8 · 45 MINUTES

Real-World IoT Applications and Project Showcase

Design a real IoT system, build the Telecom Site Monitor, and pitch your business case.



Mehdi Lotfinejad

Introduction to the Internet of Things

THE ROAD SO FAR

Eight modules, one Smart Site Monitor

PART 1 — DEVICES AND DATA

01
What is IoT?
Lecture

02
How it's built
Lecture

03
Sensors and actuators
Lab 2

04
Getting online
Lab 3

PART 2 — CLOUD, AUTOMATION AND SECURITY

05
Cloud and dashboards
Lab 4

06
Automation
Lab 5

07
Security
Group activity

08
Real world + pitch
You are here

BY THE END OF THIS MODULE

You will be able to...

01 **Use** a simple design method: problem, requirements, layers

02 **Explain** IoT choices for towers, assets and data centres

03 **Build** the Telecom Site Monitor in Lab 7

04 **Estimate** the costs and the payback of your idea

05 **Pitch** your business case in two minutes

Part A

A design method you can reuse

Problem → requirements → each layer, chosen for a reason.

THE DESIGN METHOD

Problem, requirements, then each layer

Problem

Who has it? What does it cost today?

Requirements

What must it do, where, how fast?

Layers

Device, link, edge, cloud, app.

The why

Why this, not that?

Your canvas follows the same order: box 1 is the problem, boxes 2–5 are the design.

STEP 1 · THE PROBLEM AND THE PEOPLE

“When the grid fails at night, nobody sees the battery run down until the tower site goes off air.”

Ask: what problem? Who has it? What does it cost?

Part B

Real sites, real decisions

The tower site, assets on the move, our own data centres — and a farm.



Photo: pslim, CC BY 2.0, via Wikimedia Commons

SECTOR CASE 1 · SMART TOWER MONITORING

The tower site

- Sense: battery, fuel level, door, shelter heat
- Tower **RMS** sends every site’s alarms to the **NOC**
- Energy: **15–20%** of operator costs (GSMA Intelligence, 2025)
- Fuel drops, generator off? **Possible theft**
- Battery trend, run-hours: **fix before it fails**

PAUSE AND CHECK

At a tower site, where should each rule run: on site, or in the NOC? (a) Start the generator on low battery. (b) Spot fuel theft.

Think about what happens when the link to the site is down. Thirty seconds with your team.

SECTOR CASE 2 · ASSET TRACKING

Where is it, and is it moving?



Generators

Mobile units sent where the grid fails



Batteries

Valuable, and a target for theft



Spare parts

Which store has the part we need?



Field vehicles

Which team is nearest the fault?

A small **cellular IoT tracker** (Module 4): its position, and an alarm if it moves when it shouldn’t.



Photo: Daoducquan, CC BY-SA 4.0, via Wikimedia Commons

SECTOR CASE 3 · THE TELCO'S OWN DATA CENTRES

Data centres and exchanges

- Sense: inlet air, humidity, leaks, **UPS** health
- Edge: a local alarm and extra cooling
- **DCIM** software: every rack on one screen
- Server inlet air: **18–27 °C** (ASHRAE)
- **PUE 1.56**: 0.56 kWh extra per 1 kWh of IT (Uptime, 2024)

Every choice traces back to a requirement

Layer	Choice	Why
Devices	Soil and temperature sensors; relay for the valve	Measures what matters; switches the valve safely
Connectivity	**LoRaWAN** to one farmhouse gateway	Kilometres of range, years on a battery
Edge	Open below **30%** moisture, close above **45%**	Works if the internet drops; no flickering
Cloud	Store readings; alert if a sensor is silent for 2 h	Checks across all fields, plus history
Application	Phone dashboard: one status tile per field	See what needs attention in seconds
Security	Own keys per sensor; MFA; signed updates	Hard to fake readings; can be patched



Photo: Christianjhart1, CC0, via Wikimedia Commons

THE SAME DESIGN AT PLANTATION SIZE

Smart agriculture

- Soil, weather and tank-level sensors
- **LoRaWAN** or **NB-IoT**: big farms, small messages
- Watering decided **locally**; cloud adds forecasts
- Low-power cellular IoT: **1 billion** connections by the end of 2025 (GSMA)

IN TELCO OPERATIONS

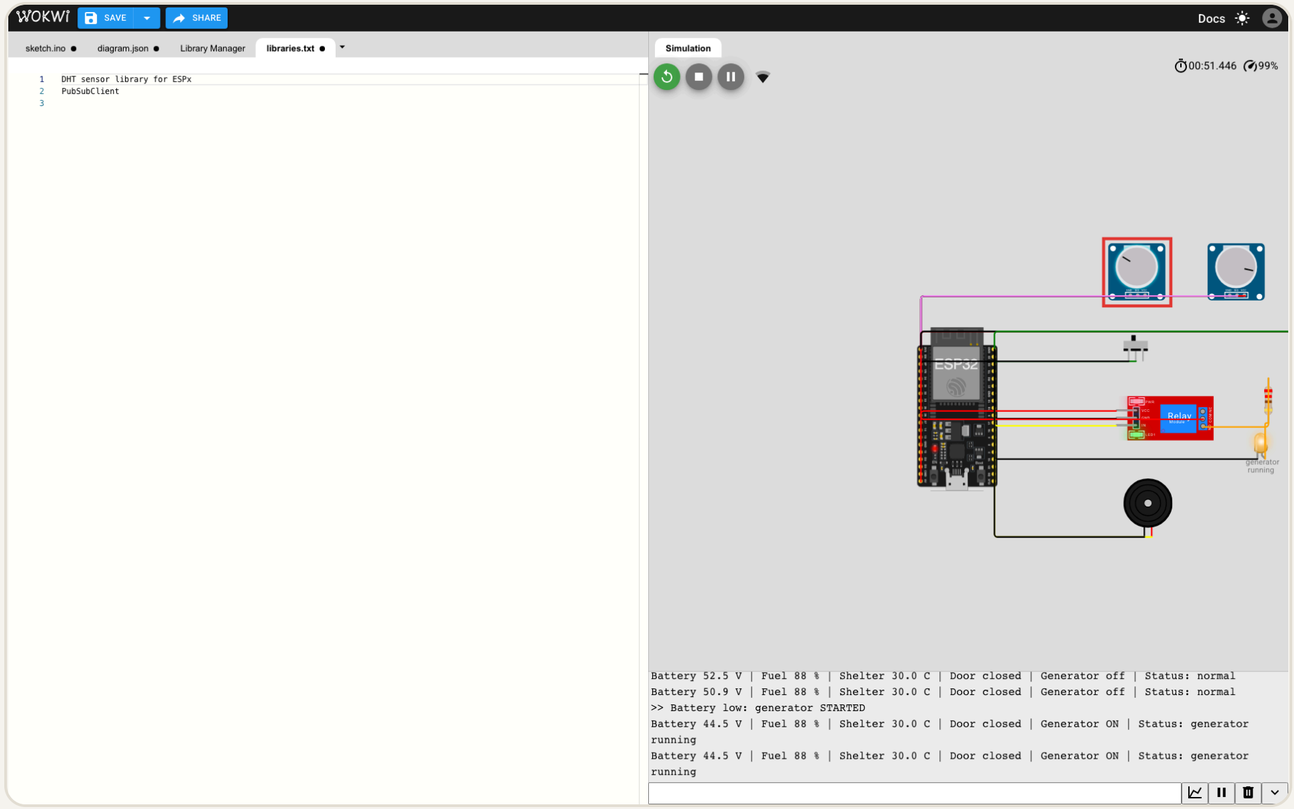
Today, with IoT, and the benefit

Operation	Today	With IoT	The benefit
Site energy and diesel	Fuel and batteries checked on visits	Live battery and fuel; generator only when needed	Less diesel, fewer site visits
Site security	Theft found at the next visit	Door and theft alarms reach the NOC	Less theft, faster action
Maintenance and spares	Repair after it fails; hunt for parts	Battery trends, run-hours, tracked spares	Faster repair, better use of engineers

No numbers yet: your team estimates them for its own case on the canvas.

LAB 7 · 18 MINUTES HANDS-ON

Capstone: the Telecom Site Monitor



1. New ESP32 project: paste sketch.ino and change **YOUR_NAME**
2. Paste diagram.json, add libraries.txt, press Play
3. Generator off: turn the **right dial** (fuel) down → theft alarm
4. Turn the **left dial** (battery) below **47 V** → the generator starts
5. Click the slide switch → door alarm; then watch it all in HiveMQ

Scan for the step-by-step guide
github.com/datatweets/intro-to-iot-workshop

BUSINESS-CASE CANVAS · BOXES 6 AND 7

Costs and payback

Devices

Sensors, boards, installation

Connectivity

A fee per device, every month

Platform

Cloud, RMS or DCIM software

People

Set-up, checks, training

7. Payback

Total cost ÷ saving per month

Box 6 is every cost, for years — not only the device price. Round numbers are fine.

THE PITCH · 2 MINUTES PER TEAM

The committee listens for three things



Clear problem

Who has it, what it costs



One number

A saving you can explain



One named risk

From box 8, with your fix

The steering committee: your trainer and, ideally, a sponsor from the business.

BUSINESS-CASE CANVAS · 13 MIN

Finish your canvas, then pitch

1. Boxes 6 and 7: list the costs, estimate the payback
2. Choose a presenter; the team holds the canvas
3. Pitch to the steering committee: 2 minutes
4. Committee: problem? number? risk? One line each
5. Vote: which pilot would you fund first?

KEY TAKEAWAYS

What to remember

- Start with **the problem and its cost**, not the technology
- Choose each layer **for a reason** you can say aloud
- Tower, assets, data centre, farm: **same five layers**
- Site data has a price: **energy, fuel, outages, visits**
- A good pitch: **clear problem, one number, one risk**

CHECK YOUR UNDERSTANDING

Quick quiz

Q1 Which two readings can reveal fuel theft at a tower site?

Q2 Why should the generator start on site, not from the NOC?

Q3 A data centre has a PUE of 1.56. What does that mean?

Q4 What three things did the steering committee listen for?

THANK YOU

Thank you — and your next steps

Take home your canvas and the 2-page manager brief. The course materials — book, labs and slides — stay available.

Mehdi Lotfinejad

Book, labs and slides: github.com/datatweets/intro-to-iot-workshop